Retained Exponents and Additive Defect Budgets for Two Recursive Tree Potentials

Dr. Muralidhara Subbarao
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Abstract

We analyze the actual least-potential norms in the assigned manuscript, rather than a Euclidean surrogate. Retaining the exponent in its accumulated exit-energy recurrence improves the variable-height estimate from power three to power e when $p_h = 1 + 1/h$. More generally, every fixed exponent schedule $p_h > 1$ with $p_h^h \le Q$, $Q \ge 2$, admits an explicit power-Q finite-head and averaged midpoint estimate. We give the full component and outer-sum proofs with completed tails, the exact height-dependent constants, and their optimal uniform envelope for this recurrence. Separately, a two-field energy inequality with quantified failure of local tightness yields an additive finite-tree error certificate. The exact version replaces the source's exponentially small logarithmic-propagation constant by a polynomial bound under the same field hypotheses. Root normalization, actual norm evaluation, field slack and omitted subtrees remain distinct. These refinements do not repeat the diamond-transfer improvements in M639 or claim a new general renorming classification. Contribution assessment: C3/5, a retained power-e estimate, an explicit exponent-schedule extension, and a source-specific additive tightness-defect certificate provide notable quantitative refinements.

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1 The source model and its boundaries

All spaces and fields here are real. On a rooted tree every node has finitely many predecessors and may have countably many children. For a finitely supported array v, and a fixed exponent $p > 1$, start with zero fields below its support and compute upwards:

$$a_s = \Big(\sum_{c \in \mathrm{ch}(s)} P_c^p \Big)^{1/p}, \qquad b_s = \Big(\sum_{c \in \mathrm{ch}(s)} Q_c^p \Big)^{1/p},$$
$$P_s(v) = \max\{a_s, v_s + b_s\}, \qquad Q_s(v) = \max\{a_s - v_s, b_s\}. \tag{1}$$

Thus $P - Q = v$. Both fields dominate their child norms and at least one of these inequalities is tight at each node. Induction proves that this is the coordinatewise least feasible nonnegative pair. Negation exchanges the fields; addition of feasible pairs and minimality prove subadditivity.

1.1 Root conventions cannot be interchanged

The Euclidean part of the source uses $p = 2$ and three conventions. On $\mathbb{N}^{<\omega}$, $N_\Sigma(v) = P_o(v) + Q_o(v)$, with completion X_Σ. On the tree joining all finite-height $T_h = \mathbb{N}^{\le h}$ to one new root ρ, the fixed-zero-root space has $v_\rho = 0$ and norm $N_0(v) = P_\rho(v) = Q_\rho(v)$. Allowing v_ρ to vary gives instead the root-sum norm $N_J(v) = P_\rho(v) + Q_\rho(v)$ and its completion X_J. On the root-zero subspace $N_J = 2N_0$.

An initial set contains the root and every predecessor of each of its nodes. Restriction Π_H to a finite initial set is contractive, by monotonicity of (1) in the child fields. Its kernel is finite codimensional and finite tails are dense there: apply $I - \Pi_H$ to finite approximations. These facts do not say that arbitrary coordinate deletions are contractive, or that a finite-dimensional Hilbert metric equals the recursive norm.

For a Banach norm N, the averaged asymptotic midpoint modulus is

$$\widehat{\delta}_N(t) = \inf_{N(x)=1} \sup_{\substack{F \text{ closed} \\ \operatorname{codim} F < \infty}} \inf_{\substack{y \in F \\ N(y) \ge 1}} \left(\frac{N(x+ty) + N(x-ty)}{2} - 1 \right). \tag{2}$$

For a fixed direction, the endpoint average is an even convex function of the radius, hence nondecreasing on $[0, \infty)$. Unit directions therefore suffice in the innermost infimum. This is a two-sided property. It must not be replaced by the one-sided AUC modulus or by a statement about equivalent norms.

1.2 Architecture and scope of the reading

The assigned source [1] was freshly read in full, including all six mathematical sections. Its foundations establish the exact recursions, completions, finite-head properties and reflexivity of the joined models. Its energy section propagates a homogeneous two-field potential and proves cubic bounds. Its obstruction section excludes AUC renormings through bounded path indicators with weakly null child increments.

The stopping section has two different probability constructions. The first uses the actual masses $w_s = P_s(x)^2 + Q_s(x)^2$, child probabilities w_c/w_s, and two absorbing states for the field deficits. Its limit order is root defect first, then the smaller-field threshold, then the product threshold, and finally the large-defect threshold. The second distributes a probability mass dominating the endpoint square fields and uses two bounded supermartingales. Selected increments must be predictable. We do not replace either law by independent child samples.

The last section changes the exponent inside a height-h component to $p_h = 1 + 1/h$, then takes an outer Hilbert sum. It already proves an accumulated-energy inequality retaining p_h^h, before replacing that power by three and its constant by $3e$. Our principal refinement concerns this last simplification and an explicit extension of the allowable exponent schedule. The source's sixth-power proof, logarithmic

argument and nonrenormability claims are retained as source results unless separately rederived below. Reading them is not a claim to have independently audited every global or historical dependency. M639 used a different, fixed-zero-root cubic input for diamond descent; that graph-transfer improvement is not new credit here.

2 General exponent schedules and the actual norms

Fix numbers $p_h > 1$, constant within each T_h, and define

$$\nu_h(v) = P_o(v) + Q_o(v), \quad E_h = (\ell_{p_h}(T_h), \nu_h), \quad X_{\boldsymbol{p}} = \Big(\bigoplus_{h\geq 1} E_h\Big)_{\ell_2}.$$

The outer norm is denoted $\|\cdot\|$. A finite head for a finite-total-support center is the union of finite ancestral sets containing its support in its nonzero components.

Lemma 2.1 (The required completion properties)**.** *For each h, ν_h is a norm and*

$$(h+1)^{-1/p_h} \|v\|_{p_h} \leq \nu_h(v) \leq 2(h+1)^{1-1/p_h} \|v\|_{p_h}. \tag{3}$$

The spaces E_h and $X_{\boldsymbol{p}}$ are reflexive, with separable duals. Finite-total-support arrays are dense. Finite coordinate projections are bounded and finite rank; ancestral ones are contractive.

Proof. With $m_s = \max(P_s, Q_s)$, the recursion gives $m_s \leq |v_s| + \|(m_c)\|_{p_h}$. Taking a level norm and iterating yields $m_o \leq \sum_{j=0}^{h} \|v|_j\|_{p_h}$. Since $\nu_h \leq 2m_o$, finite-dimensional Hölder gives the upper bound. On each level the p_h-norms of the two fields are bounded by their root values; $v = P - Q$ therefore gives the lower bound. This also defines the recursion for every $v \in \ell_{p_h}$, since there are only finitely many levels. Least-field addition proves the triangle inequality, sign exchange gives symmetry, and the lower estimate gives definiteness.

Norm equivalence gives completeness, reflexivity, dual separability and coordinate density for each component. The dual of the outer ℓ_2 sum is the ℓ_2 sum of the component duals: Cauchy–Schwarz gives one inequality and finite almost-norming component tests give the reverse. Applying this twice proves reflexivity. Truncating components and then their coordinates proves density. Coordinate evaluations are bounded; finite coordinate restrictions are finite sums of bounded rank-one maps. Ancestral contractivity follows by upward monotonicity. No uniform Hilbert comparison in h was used. □

3 Retaining the accumulated exponent

The following proof retains the source's exact exit-energy recurrence, but does not round its exponent or constant.

Theorem 3.1 (Height-dependent component estimate)**.** *Fix height $h \geq 1$ and $p > 1$. Let $x \in E_h$ be supported on a finite ancestral set A and let $y \in E_h$ vanish on A. Set*

$$B = \nu_h(x), \quad Y = \nu_h(y), \quad M_\pm = \nu_h(x \pm y), \quad M = \max(M_+, M_-), \quad q = p^h,$$

and

$$C_{h,p} = \left(p\, 3^{(p-1)/p}\right)^{(1-p^{-h})/(p-1)}. \tag{4}$$

If $M > 0$, then

$$\frac{M_+ + M_-}{2} - B \geq M \left(\frac{Y}{2C_{h,p}M}\right)^q. \tag{5}$$

The statement includes completed, not necessarily finite, tails. For $x = 0$ it holds for every y, without a head condition. If $M = 0$, both vectors vanish.

Proof. First take $x \neq 0$ and normalize $M = 1$; then $Y \leq 1$. On A, the endpoint arrays have the common nonnegative increments

$$P_s(x + \sigma y) = P_s(x) + d_s^\sigma, \quad Q_s(x + \sigma y) = Q_s(x) + d_s^\sigma, \qquad \sigma = \pm.$$

Equality of the two increments follows from the coordinate difference, and nonnegativity from upward monotonicity, since the old exterior fields are zero. In particular $M_\sigma = B + 2d_o^\sigma$. At each node of A choose one old tight field, called S at that node. Its children in this calculation use the parent's choice, not their own choices. Subtract its old equality from the endpoint inequality. For $a, d \geq 0$, $(a + d)^p - a^p \geq d^p$ and $(a + d)^p - a^p \leq p(a + d)^{p-1}d$. It follows that

$$\sum_{c \in A \cap \mathrm{ch}(s)} (d_c^\sigma)^p + \sum_{c \in \mathrm{ch}(s) \setminus A} S_c(x + \sigma y)^p \leq p S_s(x + \sigma y)^{p-1} d_s^\sigma. \tag{6}$$

Negation exchanges the exterior P, Q arrays, so summing over signs makes the exterior term independent of the parent's label.

Put

$$L_j^p = \sum_{\substack{s \in A \\ |s| = j}} \sum_{\sigma = \pm} (d_s^\sigma)^p, \qquad H_j^p = \sum_{\substack{s \in A \\ |s| = j}} \sum_{c \in \mathrm{ch}(s) \setminus A} (P_c(y)^p + Q_c(y)^p).$$

The first-exit children form an antichain. Iterating a field's child inequalities gives the antichain bound by its root pth power, first on finite subsets and then by monotone convergence. Therefore

$$\sum_{j < h} H_j^p \leq P_o(y)^p + Q_o(y)^p \leq Y^p \leq 1. \tag{7}$$

The endpoint S-fields in (6), at a fixed level and over both signs, have total pth power at most two. Define

$$K_j = \left(L_j^p + \sum_{i < j} H_i^p \right)^{1/p}, \qquad 0 \leq j \leq h.$$

Sum (6) over the level and both signs, retain the previous exit mass, and use $\sum_{i<j} H_i^p \leq p \sum_{i<j} H_i^{p-1} H_i$. Hölder on this single joint list gives

$$K_{j+1}^p \leq p\, 3^{(p-1)/p} K_j. \tag{8}$$

The first factors have total $p/(p-1)$ power at most $2 + 1$; the second factors have total p power K_j^p. It is important to retain both signs and all past exit levels inside this list.

Iteration of (8) gives $K_h \leq C_{h,p} K_0^{p^{-h}}$, with the exact finite geometric sum in (4). On A, the two fields of y agree. Their common pth power at s equals the internal-child sum plus the maximum of the two exterior-child sums. Bounding that maximum by their sum and telescoping proves $Y = 2P_o(y) \leq 2K_h$. Finally,

$$\frac{M_+ + M_-}{2} - B = d_o^+ + d_o^- \geq \left((d_o^+)^p + (d_o^-)^p \right)^{1/p} = K_0.$$

These statements prove (5) after scaling. All countable child sums are justified by the antichain bounds; no passage from finite probes to arbitrary tails is presumed. For $x = 0$, the left side is $Y = M$, while $C_{h,p} \geq 1$, so the assertion follows directly. □

Lemma 3.2 (Uniform envelope for a schedule)**.** *Let $Q \geq 2$ and suppose $p_h > 1$, $p_h^h \leq Q$ for all h. Define*

$$C_Q = (3e)^{1-1/Q}, \qquad c_Q = (2C_Q)^{-Q}. \tag{9}$$

Then $C_{h,p_h} \leq C_Q$, and (5) implies

$$\frac{M_+ + M_-}{2} - B \geq c_Q \frac{Y^Q}{M^{Q-1}}. \tag{10}$$

The constant C_Q is the supremum of the recurrence constants $C_{h,p}$ over these schedules; this is not a sharpness assertion for the Banach-space inequality.

Proof. Taking logarithms in (4) gives

$$\log C_{h,p} = (1 - p^{-h}) \left(\frac{\log p}{p-1} + \frac{\log 3}{p} \right) \leq (1 - Q^{-1})(1 + \log 3).$$

Here $\log p \leq p - 1$ and $p > 1$. The upper envelope is approached along $p = Q^{1/h}$ as $h \to \infty$. Since $0 \leq Y/M \leq 1$, $C_{h,p} \leq C_Q$ and $p^h \leq Q$,

$$\left(\frac{Y}{2C_{h,p}M} \right)^{p^h} \geq \left(\frac{Y}{2C_Q M} \right)^Q.$$

This proves the component bound. It makes no claim of extremizing vectors for the discarded Hölder or exit-mass inequalities. □

4 Outer sums, weak tails and the power-e gain

Theorem 4.1 (Power-Q finite-head estimate)**.** *Under the schedule hypothesis of Lemma 3.2, let $x \in X_{\boldsymbol{p}}$ have finite total support and let y vanish on a finite ancestral head in every nonzero component of x. If $\|x \pm y\| \leq R$, $R > 0$, then*

$$R^2 \geq \|x\|^2 + a_Q R^{2-Q} \|y\|^Q, \qquad a_Q = 2^{-Q/2} c_Q. \tag{11}$$

Proof. Use $B_h, Y_h, M_{h,\pm}, M_h$ componentwise, and $m_h = (M_{h,+} + M_{h,-})/2$. The preceding theorem, including zero-center components, and $m_h \geq M_h/2$, yield

$$\frac{M_{h,+}^2 + M_{h,-}^2}{2} - B_h^2 \geq (m_h - B_h)(m_h + B_h) \geq \frac{c_Q}{2} \frac{Y_h^Q}{M_h^{Q-2}}.$$

Terms with $M_h = 0$ are zero. For $Q > 2$, Hölder gives

$$\sum_h Y_h^2 \leq \left(\sum_{M_h > 0} \frac{Y_h^Q}{M_h^{Q-2}} \right)^{2/Q} \left(\sum_h M_h^2 \right)^{(Q-2)/Q}.$$

For $Q = 2$ the corresponding conclusion is simply equality of the numerator sum with $\sum_h Y_h^2$. Since $\sum_h M_h^2 \leq 2R^2$, summation proves (11). Finite partial sums followed by monotone convergence justify every infinite sum. □

Corollary 4.2 (Weak tails and the averaged modulus)**.** *For any $x \in X_{\boldsymbol{p}}$, a weakly null sequence satisfying $\|y_j\| \geq \varepsilon > 0$, $\|x \pm y_j\| \leq 1$, obeys*

$$\|x\|^2 \leq 1 - a_Q \varepsilon^Q.$$

For $0 < t < 1$,

$$\widehat{\delta}(t) \geq \frac{a_Q t^Q}{2(1+t)^{Q-2}(2+t)}. \tag{12}$$

Proof. For the first assertion, choose a finite-total-support x' within $\eta > 0$ of x, and let P be the finite-rank head projection in its nonzero components. Weak nullity implies $\|Py_j\| \to 0$. Apply (11) to x', $y_j - Py_j$, at radius $1 + \eta + \|Py_j\|$. First let $j \to \infty$, then $\eta \downarrow 0$.

For a finite unit center and a unit head-vanishing y, both endpoint norms are at least one, componentwise by monotonicity. If $R = \max \|x \pm ty\|$, then $1 \leq R \leq 1 + t$, and (11) gives $R^{Q-2}(R+1)(R-1) \geq a_Q t^Q$. The mean excess is at least $(R-1)/2$, proving the bound. For an arbitrary unit center use a finite unit approximation and its head; changing the center loses at most its approximation error uniformly over unit directions in that kernel. Let the error tend to zero in (2). Radius monotonicity handles directions of norm greater than one. □

Corollary 4.3 (The actual source schedule)**.** *For $p_h = 1 + 1/h$, take $Q = e$. Equations* (11) *and* (12) *hold with*

$$C_* = (3e)^{1-1/e}, \qquad c_* = (2C_*)^{-e}, \qquad a_* = 2^{-e/2} c_*.$$

These are power-e, rather than power-three, estimates in the source's actual variable-exponent norm.

Proof. $(1 + 1/h)^h \leq e$. Its limit is e, and $\log C_{h,1+1/h} \to (1 - e^{-1})(1 + \log 3)$. Apply the preceding results. □

The source's component constant is $(6e)^{-3}$. Here $C_* < 3e$ and $e < 3$, so $c_* = (2C_*)^{-e} > (6e)^{-3}$. For normalized tails of size at most one, retaining the smaller power also increases the guaranteed gain. This is a refinement of the source's displayed accumulated-energy proof, not an independent discovery of its recurrence or of midpoint convexity.

5 Two-field energy without exact local tightness

This section returns to Euclidean child aggregation. It concerns four feasible fields, not the variable-p component theorem. On a finite rooted tree let $P, Q, d \geq 0$, with all four fields $P, Q, P + d, Q + d$ satisfying the upper-field inequalities $A_s^2 \geq \sum_{c \in \mathrm{ch}(s)} A_c^2$. Define the actual squared base slacks

$$\alpha_s = P_s^2 - \sum_c P_c^2, \qquad \beta_s = Q_s^2 - \sum_c Q_c^2,$$

and the source energy

$$E_s = [d_s(2P_s + d_s)(2Q_s + d_s)]^{2/3}. \tag{13}$$

All roots and fractional powers are nonnegative real ones, including at zero.

Theorem 5.1 (Additive tightness-defect certificate)**.** *At each internal node put*

$$U_s^P = d_s(2P_s + d_s), \qquad V_s^Q = (2Q_s + d_s)^2,$$
$$\epsilon_s^P = \big[(U_s^P + \alpha_s)^{2/3} - (U_s^P)^{2/3}\big](V_s^Q)^{1/3},$$

and define ϵ_s^Q *by exchanging* P, Q *and* α, β*. Let* $\epsilon_s = \min(\epsilon_s^P, \epsilon_s^Q)$*. For any subset* $\mathcal{B}$ *of the leaves,*

$$\sum_{b\in\mathcal{B}} d_b^2 \le \sum_{b\in\mathcal{B}} E_b \le E_o + \sum_{s\ internal} \epsilon_s. \tag{14}$$

One may use the simpler bounds

$$\epsilon_s^P \le [\alpha_s(2Q_s + d_s)]^{2/3}, \qquad \epsilon_s^Q \le [\beta_s(2P_s + d_s)]^{2/3}.$$

No positive lower bound on the fields is required.

Proof. Subtracting the base square sum from the endpoint inequality gives

$$\sum_c U_c^P \le U_s^P + \alpha_s.$$

The triangle inequality for the child vectors $Q + d$ and Q gives $\sum_c V_c^Q \le V_s^Q$. Hölder, with the same child index in both factors, yields

$$\sum_c E_c = \sum_c (U_c^P)^{2/3}(V_c^Q)^{1/3} \le (U_s^P + \alpha_s)^{2/3}(V_s^Q)^{1/3} = E_s + \epsilon_s^P.$$

The exchanged calculation gives ϵ_s^Q, so one can take their minimum separately at every node. Summing these inequalities through the finite tree cancels each nonroot internal energy once. Discarding unselected leaf energies proves the second inequality. Since $2P_b + d_b \ge d_b$ and $2Q_b + d_b \ge d_b$, $E_b \ge d_b^2$, even if the base fields do not vanish there. Finally $(u+a)^{2/3} - u^{2/3} \le a^{2/3}$ for $u, a \ge 0$, by concavity or subadditivity of this power. This proves the stated simpler bounds and covers every zero endpoint. □

For exact least-field data, at least one of α_s, β_s vanishes, so $\epsilon_s = 0$. The exact source energy proof is therefore recovered. The extension is that both base inequalities may now have slack, and all losses are accounted for additively, without inserting a multiplicative constant at each tree level. The sum of losses can still grow with height or branching. A small per-node tolerance alone does not give a height-uniform certificate.

Example 5.2 (Feasibility is not minimality)**.** Take a root with one leaf, with

$$P_o = Q_o = 1, \quad P_b = Q_b = 0, \quad d_o = 0, \quad d_b = 1.$$

All four fields are upper fields. The root energy is zero but the leaf energy is one, so the inequality without a tightness correction would be false. Here $\alpha_o = \beta_o = 1$ and $\epsilon_o = 2^{2/3} \ge 1$, and (14) is valid. This is not a counterexample to the source: its exact least fields cannot have these nonminimal base values. It is a negative control for extending the source to arbitrary feasible approximations.

5.1 What an approximate computation must certify

The common increment d is part of the theorem, not an arbitrary choice of unrelated errors for the two fields. Moreover, upper feasibility must cover every child, including omitted children. If a computation retains only some children, rigorously bounded omitted square mass must be included when bounding α, β and checking endpoint feasibility. Finite positive matrices or point evaluations do not supply that remainder.

For a finite set of interval data, one can enclose the nonnegative products in (13), their fractional powers, and each ϵ_s, using monotonic interval operations before summing. The sign of the base slacks and feasibility of the four arrays must be independently certified. A computed upper pair need not be the least pair; its root cost is not automatically the original norm, and its defect need not be the

actual midpoint defect. Connecting such data to a norm estimate requires validated enclosures for the true least fields or a separate stability bound.

The theorem also has a limited infinite-tree consequence. If every finite ancestor-closed truncation has verified losses from one nonnegative summable family (ϵ_s), then all finite boundary subsets have energy at most $E_o + \sum_s \epsilon_s$. Monotone passage through increasing boundary subsets is legitimate. This statement does not prove a zero remainder at infinity, density in an unspecified completion, or convergence of a stopping algorithm. Those are separate hypotheses.

6 A polynomial replacement for logarithmic propagation

The source's uniform logarithmic-propagation lemma assumes a finite tree, four nonnegative upper fields $p, q, p + r, q + r$, one tight base inequality at each internal node, and

$$p_o + q_o + 2r_o \le M, \qquad \sum_{b \text{ leaf}} r_b^2 \ge b^2, \qquad M \ge b > 0. \tag{15}$$

It obtains a positive constant using a logarithmic transform, predictable increments and a bound of the form $(bh/8)\exp(-K)$, with $h = b^2/(2M^2)$ and K of order $h^{-1}\log(M/(b\sqrt{h}))$. The next consequence uses the same field hypotheses without that probabilistic construction.

Corollary 6.1 (Polynomial root detection)**.** *Under* (15) *and the preceding tightness assumption,*

$$b^3 \le r_o(2p_o + r_o)(2q_o + r_o) \le r_o(M - r_o)^2, \qquad r_o \ge b^3/M^2. \tag{16}$$

Necessarily $b \le M/2$. More precisely, if $\theta(u)$ is the unique root in $[0, 1/3]$ of $\theta(1-\theta)^2 = u^3$, $0 \le u \le 1/2$, then

$$r_o \ge M\theta(b/M). \tag{17}$$

For the nonminimal data of Theorem 5.1, the weaker but explicit replacement is

$$r_o \ge \frac{(b^2 - \sum_s \epsilon_s)_+^{3/2}}{M^2}.$$

Proof. In the exact case, (14) gives $b^2 \le E_o$; raising to power $3/2$ proves the first inequality. The identity

$$(2p_o + r_o)(2q_o + r_o) = (p_o + q_o + r_o)^2 - (p_o - q_o)^2 \le (M - r_o)^2$$

proves the second, and $(M - r_o)^2 \le M^2$ gives the displayed polynomial lower bound. The upper-field inequalities imply $\sum_b r_b^2 \le \min\{(p_o + r_o)^2, (q_o + r_o)^2\} \le M^2/4$. The function $s(1 - s)^2$ is strictly increasing on $[0, 1/3]$ with maximum $4/27 > 1/8$, so θ exists uniquely on the stated domain. If $r_o/M \ge 1/3$, (17) is automatic; otherwise strict increase and (16) imply it. For nonminimal fields use $(b^2 - \sum_s \epsilon_s)_+ \le E_o$ instead. □

The source energy lemma explicitly assumes zero base fields on the selected leaves to get equality $E_b = d_b^2$. Its own internal calculation, rederived here, gives $E_b \ge d_b^2$ without that leaf restriction. This is the link to the more general logarithmic lemma. The polynomial replacement is a consequence of that source energy, not a new martingale theorem or a claim that the logarithmic proof was invalid.

Example 6.2 (An exactly enclosed scalar root)**.** Take $M = 2$, $b = 1$. The simple lower bound is $r_o \ge 1/4$. The retained polynomial gives

$$r_o \ge 2\theta(1/2) = \frac{3 - \sqrt{5}}{2}.$$

An exact rational enclosure is $381/1000 < (3-\sqrt{5})/2 < 383/1000$, verified by squaring the corresponding positive rational bounds on $\sqrt{5}$. It is an enclosure for a lower certificate, not evidence that an actual tree attains equality in all inequalities.

7 Examples, original-corpus comparisons and limits

Example 7.1 (Quadratic schedules and a flat endpoint)**.** The allowed schedule $p_h = 2^{1/h}$ has $p_h^h = 2$ exactly. Thus $C_2 = \sqrt{3e}$, $c_2 = 1/(12e)$, and

$$R^2 \geq \|x\|^2 + \frac{1}{24e}\|y\|^2, \qquad \widehat{\delta}(t) \geq \frac{t^2}{48e(2+t)}.$$

This changes the norm from the source schedule, rather than proving a quadratic bound in its unchanged $p_h = 1 + 1/h$ space. At height one take a root center $x = e_o$ and a tail $y = te_c$, $0 < t \leq 1$, at one child. The exact recursion gives

$$\nu_1(x) = 1, \quad \nu_1(y) = 2t, \quad \nu_1(x+y) = 1, \quad \nu_1(x-y) = 1 + 2t.$$

One endpoint is flat while the average increases by t. Neither the mean bound nor its power implies a one-sided gain for every direction.

The source's bounded-path obstruction still has its declared mechanism for any schedule $p_h > 1$: in a fixed component, a positive path indicator has norm one, each new child coordinate has norm two, and those child coordinates are weakly null by the component's equivalence to ℓ_{p_h}. Arbitrarily high components keep the path bound uniform. Applying the source's bounded-tree lemma excludes an AUC renorming. This observation retains the lemma's full weak-null and finite-codimension premises; it is not an inference from finite norm tests alone. No new resolution of the general AMUC-versus-AUC renorming question is claimed.

Two original private chapters were freshly read. The physical-metric chapter [6] proves the distinction between a compressed residual and its omitted component, and between actual norms and coefficient norms. Its Hilbert-space formulas are not transplanted to ν_h. They motivate the explicit insistence on the recursive norm and all-child square budgets. The original enclosure chapter [7] proves exact polynomial hull identities and the separation of algebraic and model remainders. The polynomial $s(1-s)^2 - u^3$ in Corollary 6.1 can therefore be checked by exact arithmetic on an interval, but feasibility and tightness of the tree fields still need their own proof.

The later private coupling comparison [8], read in its actual theorem and proof passage, warns against replacing a shared state by separately consistent marginal data. Here it is only a comparison: the two signs, child labels, and common defect must use the same arrays in Hölder's inequality. No Boolean coupling theorem is invoked as a Banach-space result, and no private-corpus title or text was sent to public search.

7.1 Advantages, costs and possible extensions

Retaining p_h^h produces a strictly stronger small-tail power in the unchanged source variable-exponent norm. The generalized schedule displays the exact condition making this argument uniform in height. The best uniform envelope C_Q here is best only for the retained recurrence constants; it does not settle the optimal modulus or the best possible exponent for the space. If $\sup_h p_h^h = \infty$, this proof supplies no positive uniform constant or finite power. It does not prove that every such schedule fails AMUC.

The additive-slack theorem removes an exact local hypothesis and is usable with certified nonminimal field data. Its cost is an explicit sum over all internal losses and a separate connection to actual norm values. Small local errors can accumulate over many branches, and cube-root behavior near zero limits naive linear error estimates. The reciprocal form in the source has the same endpoint sensitivity; dividing by a small field without a lower bound would not repair it.

Useful further work is to improve the joint level/exit budget three, to retain its actual value along the iteration, and to establish lower examples for optimal powers. Another concrete problem is a certified implementation that encloses true least fields and omitted subtrees together, using adaptive precision near the zero-field strata. None of these computational or optimality tasks is claimed completed here.

8 Current literature and verification scope

Dilworth and coauthors introduced AMUC and studied its relation to other asymptotic properties [2]. The current primary record for Baudier's September 2026 preprint [3] describes a negative answer to the general renorming question using the Kadets–Werner construction. The latter construction is documented in [4]; it is not the present recursive-potential model. The 2026 tree-index preprint [5] concerns bi-Lipschitz invariants and ordinal graph obstructions. These references provide context, not proof certificates for this manuscript.

The primary abstracts and version records were refreshed on October 9, 2026. The publisher search record for [2] was accessible, while a direct page fetch returned HTTP 403. The external full proofs were not audited here, and no historical priority search is claimed exhaustive.

All ten inventoried assigned public text files, including every mathematical module, were freshly displayed in bounded, untruncated intervals: 1,839 lines in total. The two original private chapters were read in full, 509 lines; the later coupling comparison was read at lines 246–375, 130 lines. Included private figure PDFs and other referenced proof bodies were not read. PRE hashes were persisted before the literal displays; the final ledger gives POST hashes and every unread complement. Earlier M639 reading is not counted as fresh M643 evidence.

The owned checks use the actual least-potential recursion on small trees and distinguish exact integer/rational predicates from floating-point diagnostic tests of nonintegral powers. They check root normalizations, boundary-energy inequalities, accumulated exits, local recurrence algebra, schedule constants and declared negative controls. They are supplemental finite fixtures, not a proof of uniform infinite-dimensional assertions. The latter rest on the explicit arguments and domain conditions above. No parent or earlier-report test counts are included.

The owned run passed 1,349 exact cases and 8,715 exact assertions, including 839 actual least-field tree cases. It excluded 2,098 candidates with nonrational fields from the exact suite. Separately, 200 float64 diagnostic cases passed 480 comparisons; 80 of those cases used the unchanged source schedule at heights one through four. These numerical comparisons are not interval certificates and are not counted as exact assertions.

The contribution is assessed as C3: the exponent and schedule refinements strengthen a specified source mechanism, and the slack theorem genuinely weakens its local exactness requirement with a full error account. Classical Hölder, least-field minimality, convexity and exact polynomial arithmetic remain credited tools, not claimed new theorems.

References

[1] OpenAI, *Midpoint convexity from two recursive potentials*, September 27, 2026. Pinned public source tree. Full assigned text read; historical reference proofs not independently audited.

[2] S. J. Dilworth, D. Kutzarova, N. L. Randrianarivony, J. P. Revalski and N. V. Zhivkov, *Lenses and asymptotic midpoint uniform convexity*, J. Math. Anal. Appl. 436 (2016), 810–821. Publisher DOI.

[3] F. P. Baudier, *The Kadets–Werner modification of Bourgain–Rosenthal space is asymptotically midpoint uniformly convex*, September 18, 2026. arXiv:2609.21283v1. Abstract and version record checked, not the full proof.

[4] V. Kadets and D. Werner, *A Banach space with the Schur and the Daugavet property*, Proc. Amer. Math. Soc. 132 (2004), 1765–1773. arXiv:math/0301182v1.

[5] E. Basset, G. Lancien and A. Procházka, *On metric characterizations of tree and fragmentability indices of Banach spaces*, May 27, 2026. arXiv:2605.28311v1.

[6] M. Subbarao, original private RamCalBooks, volume 21, *A Nonorthogonal Representation Carries a Metric*, `ch01d-physical-metrics.tex`, lines 1–210. Actual metric and full-residual proofs; local access only.

[7] M. Subbarao, original private RamCalBooks, volume 16, *Between-knot constraints are continuous-time conditions*, `ch05d-clearance.tex`, lines 1–299. Exact polynomial and separately justified remainder arguments.

[8] M. Subbarao, private later comparison `RamMathUnsolved/TR-meta4-math.tex`, lines 246–375. Shared-state coupling theorem and proof inspected; only a scope comparison, not a foundational tree-norm input.